

Recommendation Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Recommendation Engine** is a data-driven system developed to provide personalized recommendations to users based on their preferences, interests, and previous interactions. The main objective of the project is to demonstrate how **Data Structures and Algorithms (DSA)** can be applied to solve real-world recommendation problems efficiently. The system stores and manages user and item information using suitable data structures such as **arrays, linked lists, hash tables, trees, graphs, and priority queues**. User-item relationships and interaction patterns are analyzed to determine the relevance of different items. Searching and sorting algorithms are used to efficiently retrieve and rank potential recommendations, while similarity-based techniques can be applied to identify items with common characteristics or user preferences. A **priority queue or heap** can be used to display the most relevant recommendations at the top. The system focuses on improving **search efficiency, ranking accuracy, and response time** while handling a growing amount of data. By integrating multiple DSA concepts into a single application, the project demonstrates their practical importance in building efficient, scalable, and user-oriented recommendation systems. The proposed system can be applied to domains such as **movies, books, products, music, or online learning resources**.

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